MECAT: A Multi-Experts Constructed Benchmark for Fine-Grained Audio Understanding Tasks

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Highlights

Motivation.

- Existing audio benchmarks fail to assess fine-grained understanding.
- Annotations are often coarse, generic, or single-perspective.
- Evaluation metrics penalize novel details while tolerating overly safe descriptions.

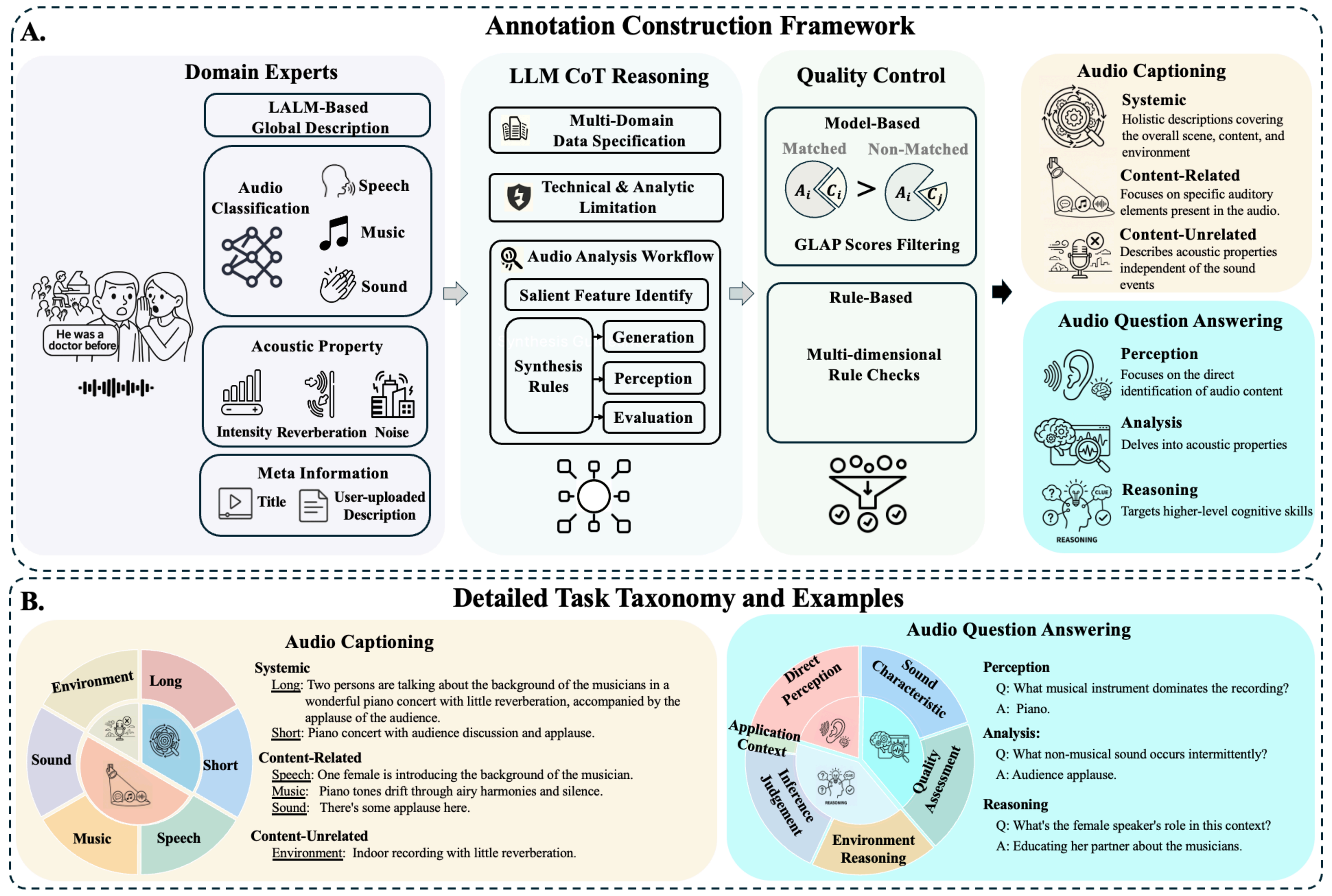
Solution.

- MECAT builds a multi-expert pipeline for detailed audio understanding.
- LLM-CoT synthesizes expert outputs into dense captions and QA pairs.
- DATE rewards specific details and penalizes generic terms.

Key advantages.

- **Scale**: 20K clips from ACAV100M across extended audio domains.
- **Density (Per Clip)**:
  - Caption: 18 refs spanning systemic, content-related/unrelated aspects.
  - QA: 5 pairs probing perception, analysis, and reasoning capabilities.
- **Discrimination**: DATE robustly separates accurate details from vague errors.

Construction Framework & Taxonomy



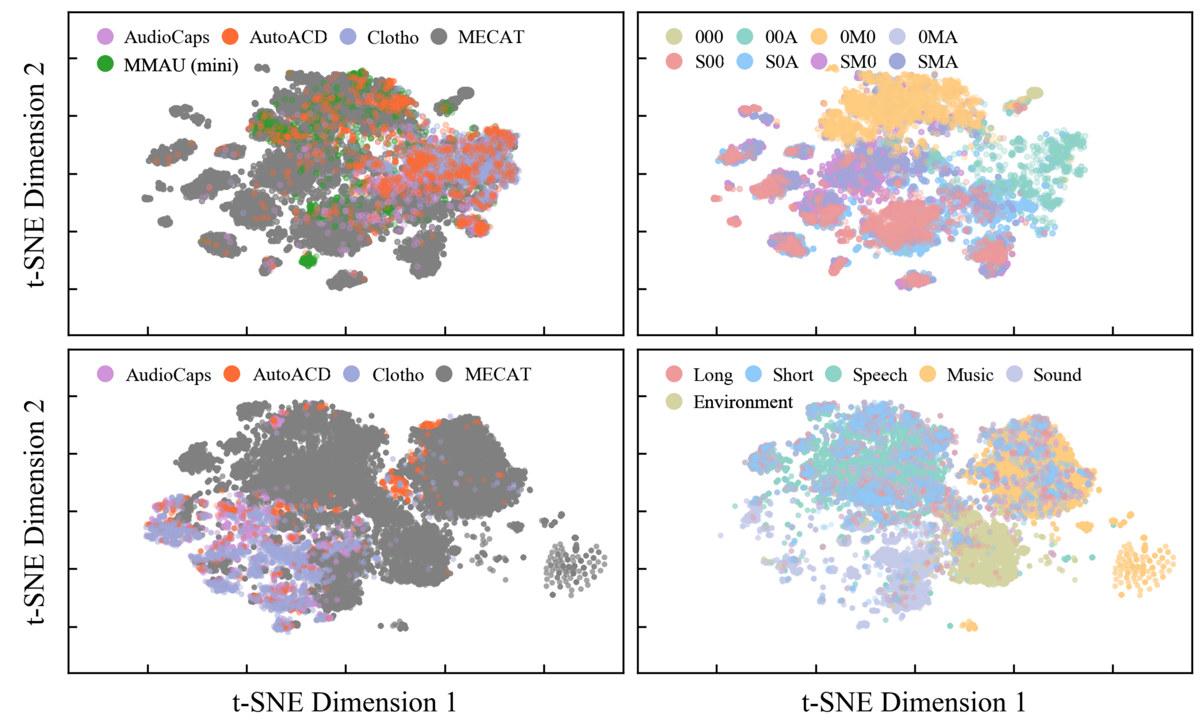
Diversity & Quality Validation

Diversity

- Top Row (Audio): Broader feature space across 8 extended domains.
- Bottom Row (Text): Richer semantic clustering across multiple categories.

Quality

- Multi-stage filtering keeps only the top 10% candidates.
- Expert audit confirms 3.81/5 relevance with only 3.4% strict error rate.



Metric Analysis

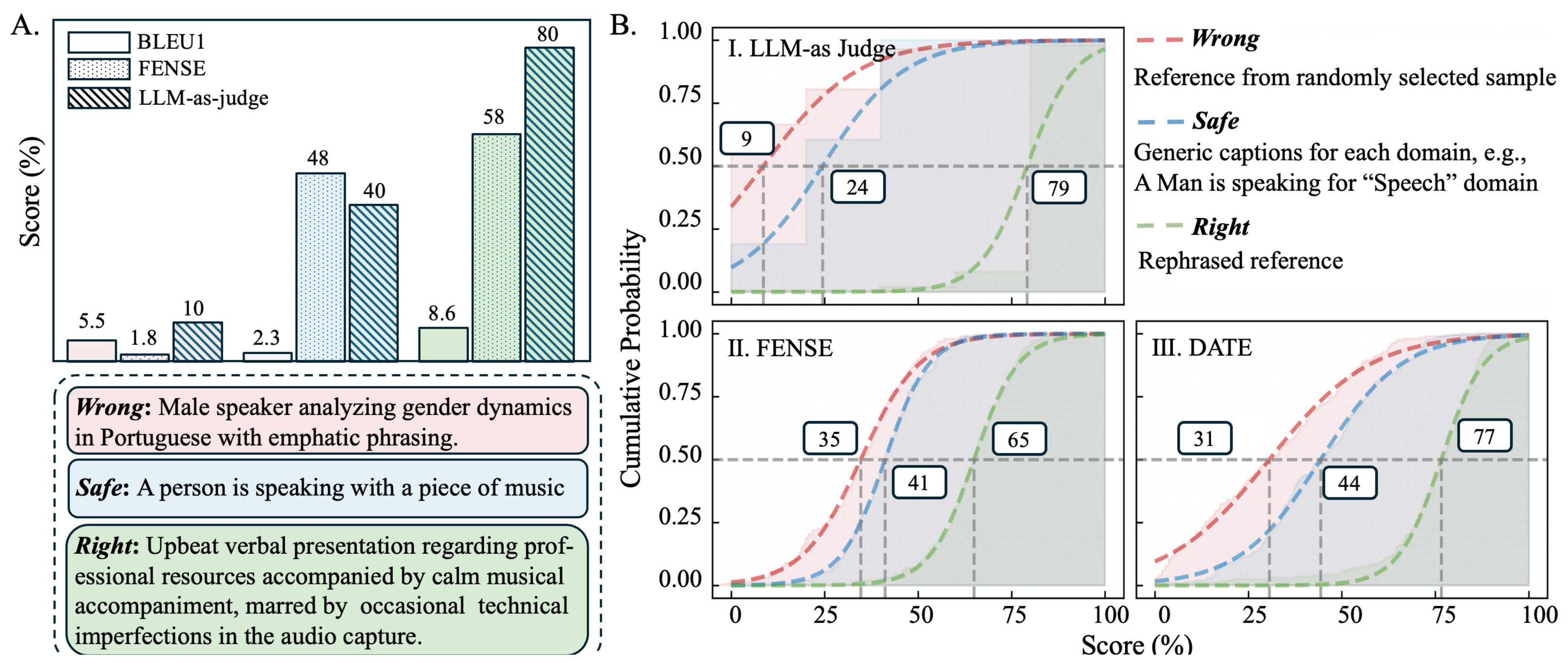
DATE penalizes generic descriptions while rewarding fine-grained accuracy.

- **Mechanism:** single-sample similarity and cross-sample discriminability.
- **Alignment:** align with human preference (preferred: 90.9 vs. non-preferred: 49.3).

$$\mathrm{DATE}_i = \frac{2 \cdot S_{sim,i} \cdot S_{dis,i}}{S_{sim,i} + S_{dis,i}} \in [0, 1]$$

Proof (Qualitative & Quantitative)

- **A**: Traditional metrics struggle to separate "Safe" from "Right".
- **B**: DATE mimics LLM-as-judge to widen the gap between "Safe" and "Right".



Benchmarking Results (Representative)

Captioning: Top LALMs improve overall scores, but

- exhibit a severe speech-centric bias
- struggle to describe content-unrelated acoustic environments.

Model	Systemic		Speech		Music		Sound		Env	$\mathrm{Score}_{\mathrm{Cap}}$
	Long	Short	Pure	Mixed	Pure	Mixed	Pure	Mixed		
Kimi-Audio	49.5	54.2	30.0	31.3	27.7	16.9	43.1	16.2	7.0	32.8
Qwen2.5-Omni	61.1	56.5	39.9	40.9	32.1	30.9	50.7	23.8	17.9	42.6
Qwen3-Omni-Flash	**65.7**	62.5	59.2	59.9	**57.4**	32.5	**55.8**	**31.6**	**27.1**	52.9
Gemini-3-Pro	64.9	**65.8**	**60.5**	**62.4**	49.8	**39.8**	51.6	30.7	26.1	**53.1**

QA: Models handle semantic questions well, but

- struggle with acoustic quality and sound-characteristic reasoning.
- fail to surpass the 40-point threshold in Quality Assessment.

Model	Direct Perception	Sound Characteristics	Quality Assessment	Inference Judgment	Application Context	$\mathrm{Score}_{\mathrm{QA}}$
Kimi-Audio	45.6	39.2	18.7	48.9	41.2	38.0
Qwen2.5-Omni	**57.8**	52.9	**39.1**	53.2	50.8	49.6
Qwen3-Omni-Flash	48.0	45.9	29.5	56.7	54.8	46.7
Gemini-3-Pro	55.5	**54.4**	37.7	**57.3**	**56.6**	**51.5**

Conclusion & Contact

Conclusion & Future work:

- **Contributions:** Dense benchmark and DATE metric for fine-grained details.
- **Insights:** Exposes LALM weakness in speech bias and mixed scenes.
- **Limitations:** 10s clip length restricts long-range temporal evaluation.
- **Future Work:** Advancing audio understanding and generation frameworks.

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